

BUILDING A NETWORKED ORGANIZATION: RESTRUCTURING THE IT DEPARTMENT AT MWH (A)

Vic Gulas caught himself, for what felt like the 10th time that day, staring at the poster board tacked on his office wall. On it were two organizational charts—old and new—for the IT department at MWH (see **Exhibit 1**). Gulas, a 20-year veteran of the engineering consulting firm, had recently been named head of IT—an entirely new role for him. His charge: to transform the department. What had been a geographically organized unit with sites in the United States, Europe, and Asia, would become a functionally organized global department that had to deliver superior, standardized service to internal and external customers—and at much lower cost.

The transformation was well under way. As the chart on the wall illustrated, the new organization, called iNet, had a new structure, which was organized around three functional areas; IT staffers were getting a feel for their new reporting relationships and new expectations. But the CEO wanted iNet fully up and running by 2004, which was just a few months away, and Gulas was worried. He knew that restructuring the IT department involved much more than moving boxes around on a chart. The key to the success of this new global department, which comprised 185 people in 27 offices in 11 countries, would be collaboration. But Gulas had a sense that many of the people who would now be expected to work together seamlessly had never even exchanged e-mails, let alone met in person. In the old organization, people who managed, say, servers in the Denver office would not have coordinated with their counterparts in Christchurch, New Zealand. Gulas himself, even after two decades at MWH, was not familiar with that many people in offices outside the United States, especially below the senior manager level. He could come up with a list of the connections between people and groups that iNet would require to deliver on its mission. But how could Gulas find out what kind of links did and did not exist already among IT employees? And how could he help people in iNet become more effectively connected to fulfill the new organization's goals? The chart he kept staring at couldn't answer those questions.

This case was prepared by McIntire School of Commerce Associate Professor Rob Cross, Vic Gulas, Amy Halliday, Andrew Parker, and Teri Vigneux. The authors would like to thank Regina Padilla and Deb Zehner for their contributions. It was written as a basis for class discussion rather than to illustrate effective or ineffective handling of an administrative situation. Copyright © 2008 by the University of Virginia Darden School Foundation, Charlottesville, VA. All rights reserved. *To order copies, send an e-mail to sales@dardenbusinesspublishing.com. No part of this publication may be reproduced, stored in a retrieval system, used in a spreadsheet, or transmitted in any form or by any means—electronic, mechanical, photocopying, recording, or otherwise—without the permission of the Darden School Foundation.*

Corporate Overview

MWH was considered a global leader in water, hydropower, and environmental remediation. The firm's engineers were frequently recognized for their excellence; for example, in 2002, they received honors from the American Academy of Environmental Engineers and the Hans Albert Einstein Award for accomplishments in sedimentation engineering.

In 2003, the organization realized more than \$975 million in revenues and employed more than 6,000 specialists. MWH's service lines included environmental engineering, power generation, facilities development, construction, multisector program management, asset management, government relations, and applied science. The firm managed hundreds of projects each year, providing services to a wide array of institutions worldwide, including municipalities, government agencies, multinational companies, industrial concerns, and military organizations. The scope of these projects was substantial. For example, MWH designed, built, financed, and managed the Karaji Dam, which supplied fresh water to the city of Tehran; the Derbendi Khan Dam in Iraq, which at the time was the highest rock-fill dam in the world; and the 5 de Noviembre Hydroelectric Project in El Salvador, which included the first large underground power plant in the Western Hemisphere. In 2003, MWH was managing the design and implementation of a capital improvement program to expand the water and wastewater infrastructure serving 7 million residential and 200,000 business customers in northwest England. This five-year, \$3.3 billion endeavor was the largest water industry program in Europe. MWH was also helping the U.S. National Aeronautics and Space Administration decommission, decontaminate, and dismantle a former research reactor site at the Plum Brook Station Reactor Facility in Sandusky, Ohio.

MWH was well known for developing and applying advanced technology to projects; it was considered an industry leader in several IT applications, including proprietary software, process automation, three-dimensional CAD, and enterprise solutions. For example, the firm applied 3-D CAD technology to design the expansion of a wastewater treatment plant in Utah. The 3-D modeling allowed the customer to visualize the proposed design and make changes directly to the model, which sped up the approval process and eliminated the need for multiple, expensive design reviews. It also helped identify conflicts that might have led to costly changes during construction.

Although MWH's projects spanned the globe, the firm did not operate as a truly global enterprise, a shortcoming the CEO was eager to rectify. Over the previous 15 years, MWH had grown through numerous mergers, some large and some small, and had not yet established consistent processes across the organization. As Gulas saw it, the firm's structure was tribal, based on business units and regional autonomy—not global—and iNet was to be the first function to establish global processes. The hope was that it would serve as a model for the globalization of other functions within MWH, such as human resources and finance.

The culture at MWH was all about relationships: Because employees had no strong process base to guide them, they got their work done by turning to other people for help. As

Gulas knew from his own experience, building trusting relationships took time. In fact, his most valuable ties, the ones he relied on time and again to get work done, had developed over many years. One was with someone he had hired a few years back. Another was with someone who had started as his temp—12 years ago—and had risen through the organization. Yet another was with someone he had gotten to know on a project designing a knowledge portal. Relationships couldn't be built in a day, but once they existed, they could be the key to getting things done quickly and effectively. This was why Gulas was eager to find out what connections did—and did not—exist in the new iNet.

Creating a Global Structure

An IT department organized by geography had been acceptable when MWH was smaller. Now this structure—which, for instance, had allowed for simultaneous use of six different e-mail systems and three enterprise resource planning (ERP) systems—made it impossible for the IT department to deliver sophisticated, standardized solutions. Although there had been some progress in standardizing major technology platforms across the company—the department had migrated the entire firm to one e-mail system, established software standards, implemented a knowledge portal, and mapped the initial steps toward creating a global ERP system—there had been only limited collaboration across the IT business centers. Best practices were, at times, shared across departmental units, but rarely adopted in any systematic way. The leaders of the various IT groups met two or three times a year in an effort to drive standard approaches for the company, but the groups continued to operate independently, each serving its own master and employing its own measures and processes. For instance, the team in Pasadena, California, which was responsible for upgrading the ERP system used in the Americas (the system that, it was hoped, would eventually be robust enough to serve all of MWH), worked in a virtual silo because it reported to the finance department, which set priorities and metrics for its work and had no incentive to collaborate with the rest of the IT organization. The group dealing with innovative design efforts was small and did some interesting things, but people there acted more like cowboys than like members of a global team. And, in fact, the department was not a global team: This design group's ability to scale its technology depended on persuading one operation at a time.

As a result of this fragmentation, the IT organization's internal customers experienced varying levels of service, functionality, and costs for technical solutions and applications. Even more crucial was service to external customers. Market trends suggested that IT would become an increasingly substantial component of MWH's revenue, as customers demanded more complex solutions and real-time collaboration tools. Furthermore, the company anticipated rapid growth in non-U.S. revenue, as increasingly advanced technical solutions were applied to new global markets. To be competitive in both the quality and cost of solutions, the IT organization had to improve its ability to transfer technical solutions and expertise on a more standard basis around the world.

Unfortunately, standardization was only part of the challenge Gulas faced. External benchmarking studies showed that the total cost of IT at MWH was higher than industry averages. As he stepped into his new role, senior management told Gulas that the IT department was costing the company too much money, and that it had been slow to develop the infrastructure necessary to meet changing market demands. The executives demanded more creativity and collaboration—and of course they needed it rapidly and at low cost.

Gulas knew that to achieve these two ambitious goals, MWH's IT function needed a unified structure. The organization designed to achieve that goal, iNet, would comprise three functional areas (see **Exhibit 1**):

- *Global service provider*: This function would provide basic IT services, including messaging, network management, asset management, and end-user support, to all MWH business units worldwide. Each of these areas would be served by a global team, with members needing to work together from their bases in Denver, Colorado, and Pasadena, California, in the United States; High Wycombe and Warrington in the United Kingdom; and Christchurch in New Zealand, where labor rates were significantly lower. In fact, the New Zealand location, which would be a global service center for commodity-type functions such as end-user support and the management of networks and servers, was one of iNet's major cost-reduction initiatives.
- *Solution development*: This function would develop, support, and maintain enterprise applications for areas such as finance, human resources, knowledge management, and engineering design. Developing and implementing enterprise applications was not easy. First, it was a challenge to convince business unit managers that their development work could be performed at significantly lower labor rates while maintaining the same quality standards. Second, the company needed to improve coordination, communication, and project management skills to offset geographical and cultural differences. As another crucial cost-reduction move, iNet would establish a center in Pune, India, to provide resources to developers—the box labeled “Resource Net” on the iNet organizational chart.
- *Strategy and administration*: Finally, the strategy and administration function would provide direction and ensure integration among iNet employees. In addition to strategy-definition and standard-setting, this group included a project management office charged with establishing and monitoring compliance with project management methodologies; managing very large, complex projects; and administering and dispensing fees for global IT services.

In iNet, Gulas would have six direct reports: three directors of the functional groups plus a director for each region (the Americas, Europe, and Asia), who would handle such field operations as client service management, regional support, and human resource issues. Gulas had conducted a lengthy interview process to identify new leadership from existing staff. Two of the three directors were located in Denver; the global services director was located in the United

Kingdom. Each regional IT director was with an operating unit in Denver, London, and Christchurch.

Of iNet's 185 employees, roughly 105 were in the United States, with primary resource centers in Denver, Pasadena, and Chicago. Average tenure was two to five years, with slightly higher turnover in the United States. As part of the restructuring, 12 positions were moved from Europe and the Americas to New Zealand; four people chose to move to the New Zealand service center. No offices were closed.

Creating the iNet structure had taken careful thought and planning. Gulas and his colleagues had gone back and forth with various designs over many months, letting their ideas percolate and trying to find the structure that could best achieve iNet's business goals. But Gulas knew that putting the new organization in place was in some ways just an initial step. The new structure couldn't accomplish anything without the right relationships.

Gulas saw the issue in both general and specific terms. Generally, the restructuring would require iNet employees to align with a given function and begin to collaborate well with people in offices around the world as well as across functional lines. And all this collaboration would need to be effective and virtually transparent to end users despite the physical distance and time zones separating these groups. Gulas thought that one good way to help bring people together would be to establish "expertise groups" that helped disseminate knowledge and best practices among those engaged in the same kind of work. After all, each IT area represented distinct technologies, expertise, and processes. To maintain world-class status in specific domains, each area would have to become a tighter unit, through which information flowed easily and evenly. In part, this would happen because of changes in reporting structure. For example, the server, network, and desktop systems area, which had previously reported to three different regional directors, would now report to the head of the Global Service Provider function. This particular group would certainly need to coordinate its expertise to fulfill a major iNet initiative: reducing the number of servers in the United States by consolidating information on larger, more robust networks. Simply changing reporting structures would not be enough to get people to work together.

In addition to considering the general connectivity and collaboration that would need to exist, Gulas was also thinking more specifically about groups that needed to be able to work together to fulfill iNet's business goals (see **Appendix**). The New Zealand global service center couldn't play its role in the global services unit unless the staffers in Christchurch had productive ties with their colleagues throughout iNet, particularly with iNet headquarters in Denver and with the global services teams in Pasadena, High Wycombe, and Warrington. Gulas was also thinking about the people who were moving to New Zealand from other locations. To create a strong group there, the newcomers would need to build new ties and learn to let go of some old ones, which would be painful. Then there was the resource net in Pune, India. That group also needed strong ties to other iNet sites, particularly to headquarters in Denver. Finally, Gulas was concerned about the team in Pasadena that was responsible for upgrading the ERP system that

would eventually serve the entire firm. Everyone there would need strong ties throughout iNet, particularly with people in New Zealand, to ensure the seamless globalization of the system.

Gulas wanted to be optimistic about how well people would adjust to their new roles and new colleagues—he wanted to think that people would reach out immediately to build the ties they would need to do their jobs. But, honestly, he knew there would be some bumps along the way. Not everyone would be on board and trust that the reorganization would succeed. He had heard some grumbling, for example, from the Walnut Creek, California, site.

A Relational View of iNet

As a senior manager, Gulas knew it was his job to orchestrate the work of others. Because much of the firm's most important work was accomplished through relationships—and not necessarily those depicted on the formal organizational chart—Gulas's job was, in effect, to orchestrate what he could not see. iNet's organizational chart couldn't show him the ties that did and did not exist within the group. He needed a view through a different lens, and he had a hunch what that lens might be.

Around the time that Gulas took on the challenge of building a global IT function for MWH, he attended a conference held by IBM's Institute for Knowledge-Based Organizations, a knowledge management think tank. One of the conference presentations was on analyzing informal networks in organizations. As the speaker explained, the idea of mapping relationships between people in groups or between entire groups was not new—researchers in sociology, anthropology, and social psychology had long studied the effects and importance of social networks—but only in the past decade or so had the field of management picked up on the idea. The speaker described in compelling terms the power of social network analysis applied to business organizations (called organizational network analysis, or ONA). Gulas was intrigued by the notion that you could ask people in an organization a few simple questions—Whom do you turn to regularly for help to do your job? Whom do you trust? Which people make you feel motivated and inspired?—plug in the responses to those questions, and come up with a picture of relationships across a group. The speaker also displayed some examples of network maps, which showed clusters of dots, representing people, connected—or not—by lines.

After listening to the conference presentation, Gulas decided that iNet would be a perfect test site for this fledgling technique. Frankly, he thought to himself, it's not like there were any intriguing alternatives to help him foster collaboration across his organization. And certainly none that would yield actual graphic representations of the organization, in addition to tables of raw data. He could spend a lot of time traveling to iNet sites, talking to people about how things were going, but that approach would yield mostly anecdotes and would eat up a lot of Gulas's time. Anyway, he had always enjoyed trying out new ideas, and when better to test something new than in the face of a novel challenge?

When Gulas decided to conduct an ONA of iNet, he felt a bit like a new parent photographing his baby, hoping to capture a certain stage of infancy. In his case, however, the result of the analysis would be not a portrait, but an X-ray. In particular, he thought that an ONA would provide a baseline view of collaboration at the birth of the new organization—a picture of who was and was not collaborating with whom. Surely this X-ray would reveal risks and potential trouble spots, but perhaps it would also identify people who already had strong ties across geographies and functions and could serve as change agents for iNet. Gulas also hoped that having this view of the inner workings of iNet would help convince others in the organization that there were collaboration issues they all needed to address.

Gulas initiated the ONA project by designing a survey and sending it to all 185 iNet employees in August 2003. The survey contained questions about which people the respondents turned to for important information to do their jobs. It also included questions about respondents' level in the organization and tenure at MWH, both of which could influence patterns of collaboration. When Gulas sent the survey to iNet employees, he included a brief memo describing the objectives of the survey, instructions on how to complete it, and an assurance of confidentiality. Employees were given 10 business days to respond and received two follow-up reminders during that time. Gulas was extremely pleased with the 90% response rate.

Even though Gulas had certainly anticipated that the analysis would demonstrate some fragmentation of iNet across geographic locations, he was still struck by the vividness of the network maps that were generated using the survey data. At first, what Gulas saw in the map of iNet's information flow was a flurry of nodes, representing people, and a dense web of lines connecting them (see **Exhibit 2**). But very quickly he saw that in fact the nodes were tightly clustered. The ERP team in Pasadena and the Resource Net in Pune looked so isolated, with only a few arrows reaching in and out. And the European, American, and Asian teams were likewise separated from one another. In short, what Gulas saw initially in the network results were what he had anticipated: The information-seeking relationships that existed at the birth of iNet were based on shared geography, not on shared functional areas. This was the overarching theme, and upon closer study, Gulas identified several unanticipated themes that he knew he would need to address for iNet to succeed.

A Sparse and Fragmented Information Network

If everyone in iNet were connected to everyone else, 100% of possible connections would exist. In fact, Gulas learned from the survey results that only 6% of connections actually existed in this group. And as **Exhibit 2** shows, these connections were not distributed evenly but were concentrated within geographic locations.

When network diagrams include more than 30 or 40 people, they can become difficult for the human eye to interpret. In the map of information flow throughout iNet, Gulas could see the large pattern of a network fragmented by geography, but to examine the inner workings of iNet in finer detail, he needed either to look at a subset of the larger group or at network information

presented in another format. **Exhibit 3** does just that, showing the percentage of information relationships that existed within and between each iNet location. If everybody in Denver were collaborating with one another, for instance, we would see 100% in the top left cell of that table. In fact, only 20% of connections existed. Gulas clearly did not want to see high numbers in every cell—100% connectivity would be unnecessary and a drag on the organization—but he did want to see specific points of integration where iNet’s strategy demanded it. A substantial problem seemed to exist with Christchurch, New Zealand, and Pune, India. The ONA revealed that these two sites enjoyed the lowest levels of collaboration with all other geographical locations. This was a particularly troubling result: If these two groups were intended to be important centers for MWH, then some direct intervention was needed.

Of course, percentages of connectivity were affected by group size. In a group of 10, one person needs to be connected with only nine others. In a group of 100, one person needs to consider relationships with 99 colleagues! In a table such as that in **Exhibit 3**, one must keep in mind group size. But the diagonal of this table, which gives percentages for internal connectivity, reveals that the highest level of internal connectivity existed at Walnut Creek (48%), a group that primarily performed asset management and procurement services for business units in the United States. While this was positive for that location internally, it raised concerns about its external connectivity. The group’s relatively low level of collaboration with others even in the same country suggested an unhealthy degree of isolation and overreliance on themselves. Gulas had known that this group was not totally on board with the reorganization plan. Now the ONA results reinforced his concerns about the group’s ability to be a team player in the iNet.

Similarly, there was only minimal collaboration between Denver and Pasadena. Each location was home to an application development group, both of which served the same customers. It had become apparent that these groups were not providing optimal solutions to internal customers because they did not leverage each other’s expertise. Furthermore, there were times when work was duplicated because one group was not aware of what the other had already done. For example, when the HR department requested a new program to process employee address changes, the Denver group created a database application and workflow, unaware that the ERP application managed by the Pasadena group already provided a solution for this task. The Pasadena office also appeared to be quite isolated, which was a major problem if it was to create an ERP system that could eventually serve the needs of the entire organization.

Gulas was also disappointed to see that the expertise groups he had envisioned were not as connected as they needed to be. The network map of information flow within the Server, Network, and Desktop Systems group, for instance, was fragmented, held together by just a few people in the United States (see **Exhibit 4**). In a group that relied on certain individuals as its “glue,” assimilating new players would be difficult.

The expertise groups represented an important lever for developing and maintaining high-end expertise within the iNet group; however, connectivity within these groups was not yet what it could be, and this would affect both the quality of solutions as well as efficiency. The solution was not immediately obvious, though. The right pattern of connectivity for each group

was dictated by the expertise in that network as well as the work of the specific group. Simply imposing a mandate that forced increased connectivity in each group was not the best solution, because this would just consume people's time and energy. But developing a customized approach for each group was also a difficult and inefficient process. Several obstacles, such as varying levels of expertise, length of service, and physical dispersion would make it difficult for people to see how their work fit together. Moreover, cultural differences, including the real and perceived dominance of the United States in setting standards, making decisions, and controlling funding, could present additional challenges.

Network Rigidity

The ONA also revealed other network concerns that Gulas hadn't thought much about before. One was hierarchy. Overall, there was a relatively high level of collaboration among the most senior iNet employees, but less connectivity was evident among those at lower levels. As Gulas studied a table that assessed connectivity within and across organizational levels (see **Exhibit 5**), he noted that hierarchy did not appear to be constraining collaboration: People did reach up in the hierarchy for information they needed to do their work. To Gulas, this was crucial: His hope for iNet's future was that more seasoned employees would transfer their knowledge and serve as mentors to those with less experience. But it troubled him to see the relative lack of connectivity among those on the front lines. These were the people who most needed to share expertise to solve client problems. Also, if junior people or new hires felt isolated from their colleagues, they were not likely to be very happy at MWH.

Another potential concern was network size. The analysis revealed that, on average, people had nine working relationships each. But the range in size of each person's network was dramatic. On the low end, several people had only one or two relationships, whereas on the high end, some people were overloaded with more than 25 people coming to them on a frequent basis for information (see **Exhibit 6**). In fact, on reviewing the analysis, Gulas remembered that one such person had just pulled three all-nighters. This employee seemed to love his work as head of a network team and was heavily relied upon by people in the group because of his expertise. But Gulas wondered how long someone could keep up a pace like that, and, more important, whether excessive reliance on a few people created a bottleneck and weakened the network. What would happen if he left MWH?

In looking at people's positions within the network, it was no surprise to Gulas that another person, Louise Martin, ranked as a highly central person. She was a technical expert in the area of networks and servers and was also seen by the more junior members as a mentor; she often "sat in" for her director when that person was away. But Gulas was concerned because Martin would not play either a manager or director's role in the reorganized department. Knowing she could be a significant change agent in the new organization, Gulas wondered how to help her find a role that would leverage her strong skills and relationships. Alternatively, the question had to be asked: Did Martin's management style empower her people sufficiently to make decisions, or did it create too much reliance on her?

Of course, at the other end of the spectrum, the highly peripheral people were also worrisome because they represented untapped expertise. Various studies had shown that more peripheral people are less satisfied in their work and more likely to leave an organization than their better-connected counterparts. As a result, Gulas was eager to locate these people and help improve their connections within the network. One example of a peripheral person was Kathy Davidson, who was the only person supporting the ERP application in Chicago—most of her peers were in Pasadena. The Chicago group had been part of an organization that was acquired by MWH a few years back. Since the merger, Davidson's boss and some of her peers had left the organization. Gulas thought the best way to help Davidson was to give the network maps to her new supervisor, Dan Carlisle. He was not at all surprised by what he saw. Davidson had recently told him that she wanted to become more integrated with the ERP group, and he had already changed her reporting relationship so that she would report directly to a manager in Pasadena. But looking at how peripheral Davidson was on the network map, Carlisle wondered if that would be enough to bring her into the team.

Gulas thought that investing some time and effort to help central and peripheral people would accomplish a lot for iNet. But what exactly could he do? Helping someone who has become overly central in a network was a markedly different challenge from drawing in someone from the periphery.

Information Biases

There also appeared to be two potential learning biases in the network. First, when asked to identify all people important to them from an informational perspective, many iNet employees listed only other members of their immediate team. Few had any ties outside of MWH (see **Exhibit 7a**), which made Gulas wonder if enough new information was entering the group to keep employees abreast of industry, technical, and market trends.

Second, when Gulas looked at the nature of interactions between colleagues, he was surprised that most were unplanned (see **Exhibit 7b**). While serendipitous interactions could promote flexibility, he wondered if it might preclude effective information-seeking. Did people get locked into a specific subset of colleagues they had come to know personally rather than reaching out to those with the most relevant expertise? How would the group successfully implement complex projects on a global scale when most information flows were so informal?

Personal Connectivity

Finally, Gulas was intrigued by the role his own personal network would play in helping him execute this change program. Of course, after 20 years at MWH, he knew many people throughout the company, but he decided to map only the people he relied on most within iNet—those who provided him with critical information he needed to make strategic decisions (see **Exhibit 8**). Not surprisingly, he saw imbalances in the makeup of this group. For instance, of the

seven people, four were in the Denver office, and four were at the director level. Clearly, Gulas would need to make a point of building ties in other iNet locations and getting to know people lower down in the hierarchy in order to extend his personal expertise and capabilities.

Next Steps at MWH

Before Gulas had decided to conduct the ONA, he had considered some across-the-board moves for boosting collaboration throughout iNet. Certainly there were lots of technologies to choose from that promised to make it easier for everyone, everywhere, to be in touch. And he knew that it never hurt to bring people face to face, so he had looked at some possible off-sites for the entire group. But now that he had this highly detailed picture of connectivity within iNet, he knew that he had been right to hold off. Collaboration had a cost, whether time or money. Simply demanding that everyone, no matter who they were, communicate more and attend more meetings might accomplish nothing. And it certainly wouldn't do anything for Gulas's popularity.

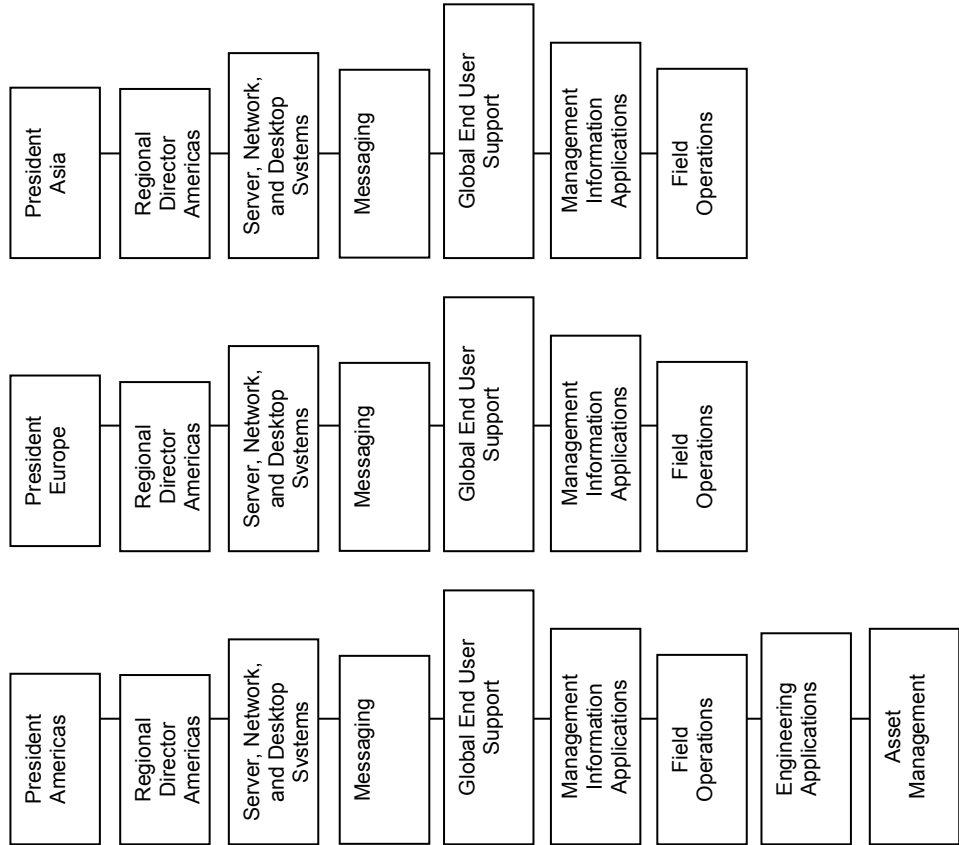
But knowing what he *shouldn't* do was one thing; knowing what he *should* do was still a big question. Instead of staring at an organizational chart, he was now busy studying the X-ray of his infant organization. This picture contained many messages, and he wasn't sure which ones to listen to first. Which gaps were the most critical? Which existing ties needed to be protected? What steps did Gulas need to take to make sure that the next X-ray of the developing iNet would reveal strong, appropriate connectivity where the organization, and all of MWH, most needed it?

Exhibit 1

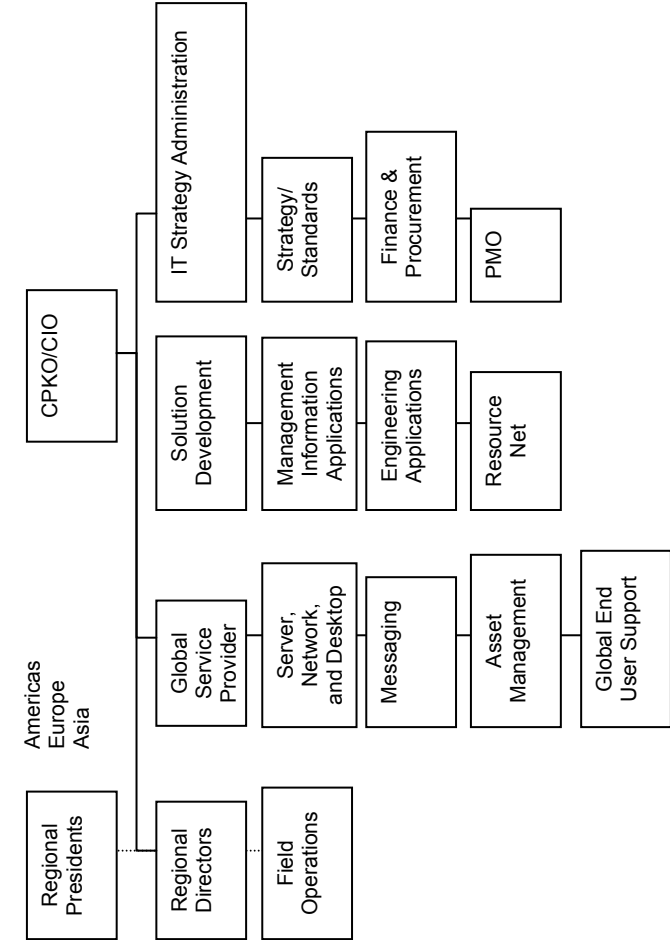
**BUILDING A NETWORKED ORGANIZATION:
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Old and New Organizational Charts for IT Department

Old Organization



New Organization

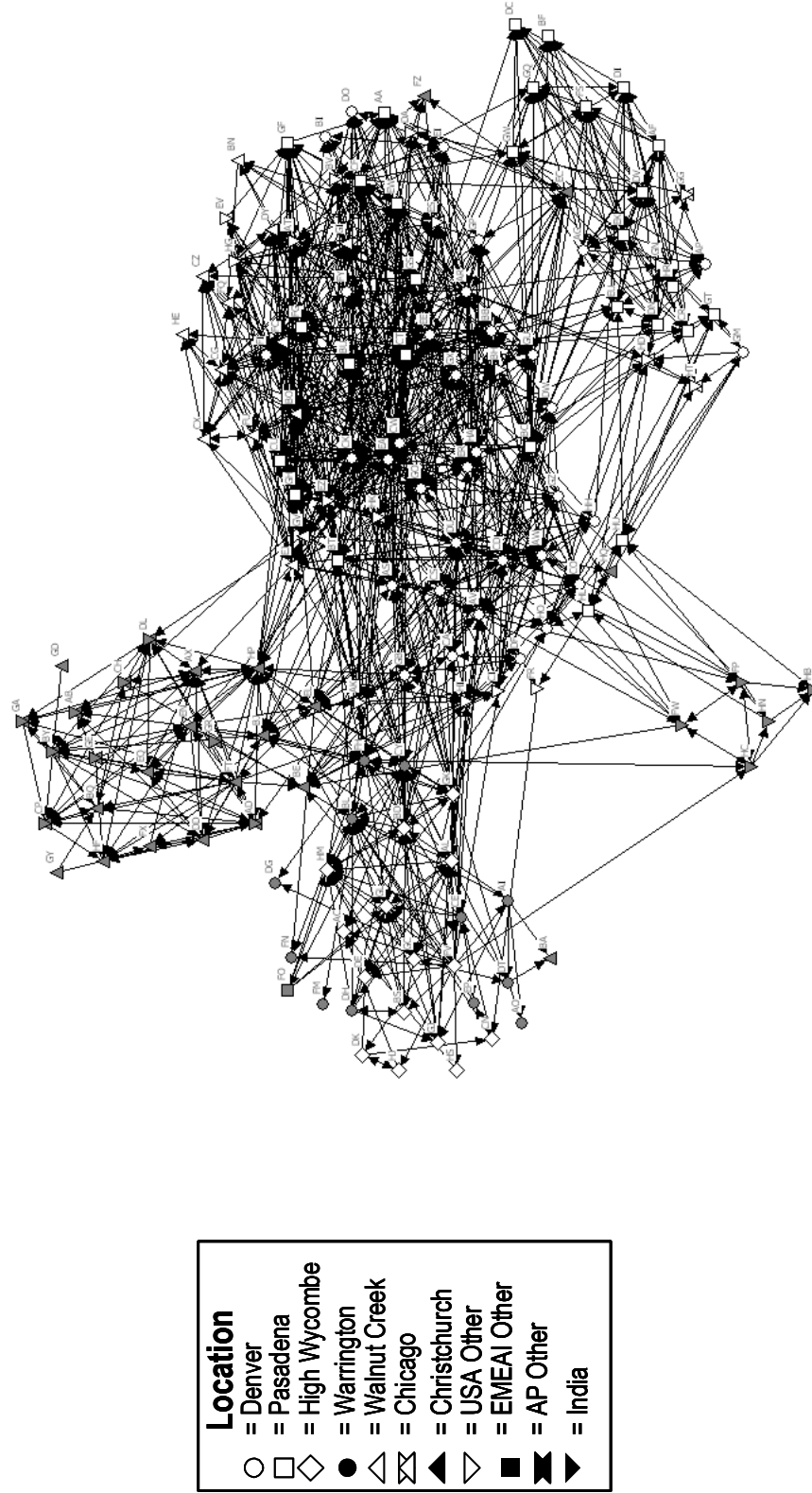


Source: Created by case writer.

Exhibit 2

**BUILDING A NETWORKED ORGANIZATION:
RESTRUCTURING THE IT DEPARTMENT AT MWH (A)**

Information Flow in the iNet Group at MWH



Source: Created by case writer.

Exhibit 3

**BUILDING A NETWORKED ORGANIZATION:
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Percentage of Information-Seeking Connectivity Within and Across Locations 2003

		Denver (34)	Pasadena (28)	High Wycombe (15)	Warrington (12)	Walnut Creek (13)	Chicago (11)	Christchurch (11)	USA (6)	EMEI (1)	AP (9)	India (5)
1	Denver	20%	6%	2%	1%	10%	12%	5%	10%	0%	0%	5%
2	Pasadena	8%	24%	0%	1%	4%	7%	2%	5%	0%	0%	1%
3	High Wycombe	3%	0%	24%	16%	0%	0%	3%	2%	0%	0%	1%
4	Warrington	1%	1%	13%	18%	1%	2%	2%	1%	8%	0%	2%
5	Walnut Creek	8%	5%	1%	0%	48%	4%	1%	1%	0%	1%	0%
6	Chicago	10%	7%	1%	2%	6%	20%	2%	3%	0%	0%	0%
7	Christchurch	2%	2%	2%	5%	1%	2%	38%	6%	0%	20%	0%
8	USA Other	9%	7%	2%	0%	4%	6%	5%	3%	0%	0%	0%
9	EMEI Other	0%	0%	20%	17%	0%	0%	0%	0%		0%	0%
10	AP Other	0%	0%	0%	1%	0%	1%	26%	0%	0%	22%	0%
11	India	2%	2%	0%	2%	0%	0%	2%	0%	0%	0%	55%

Source: Created by case writer.

Exhibit 4



Exhibit 5

**BUILDING A NETWORKED ORGANIZATION:
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Percentage of Connectivity Within and Across Hierarchies 2003

	Indiv. Contrib.	Supervisor	Project Manager	Manager	Director	Other
Individual Contributor	4%	6%	6%	6%	2%	6%
Supervisor	5%	9%	8%	10%	7%	8%
Project Manager	6%	9%	12%	11%	10%	7%
Manager	6%	10%	11%	15%	18%	12%
Director	3%	8%	12%	26%	67%	10%
Other	4%	7%	6%	7%	1%	5%

Source: Created by case writer.

Exhibit 7a

**BUILDING A NETWORKED ORGANIZATION:
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Lack of Reaching Outside of the Network

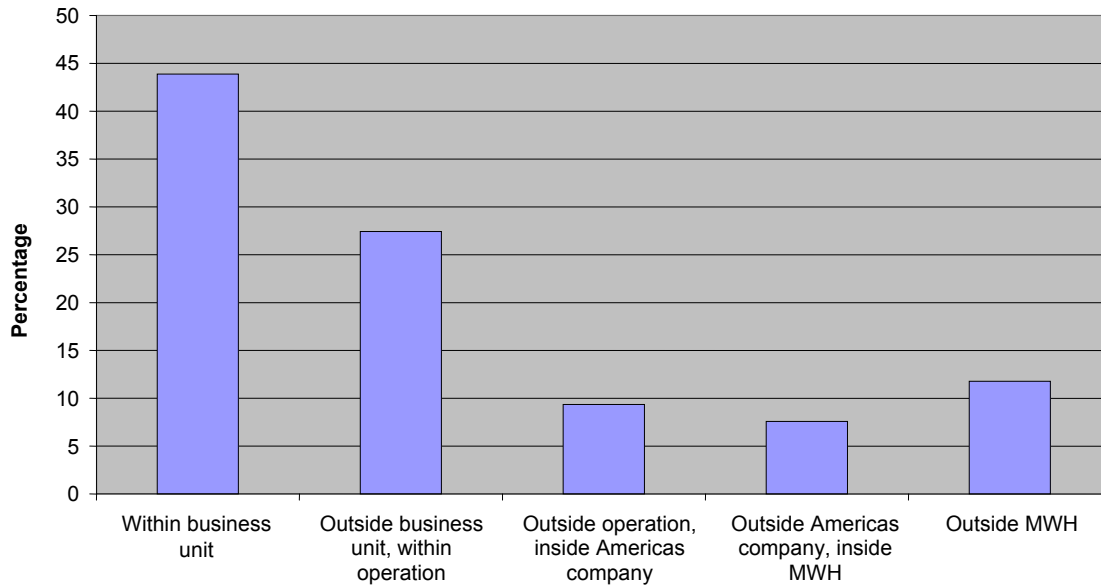
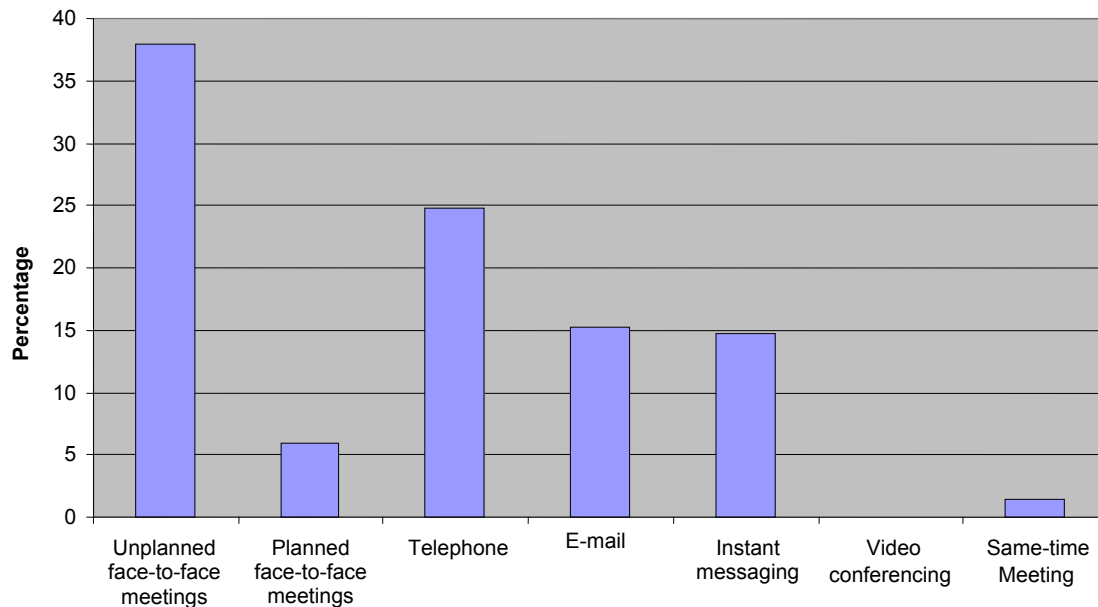


Exhibit 7b

Serendipity in Collaboration and Innovation

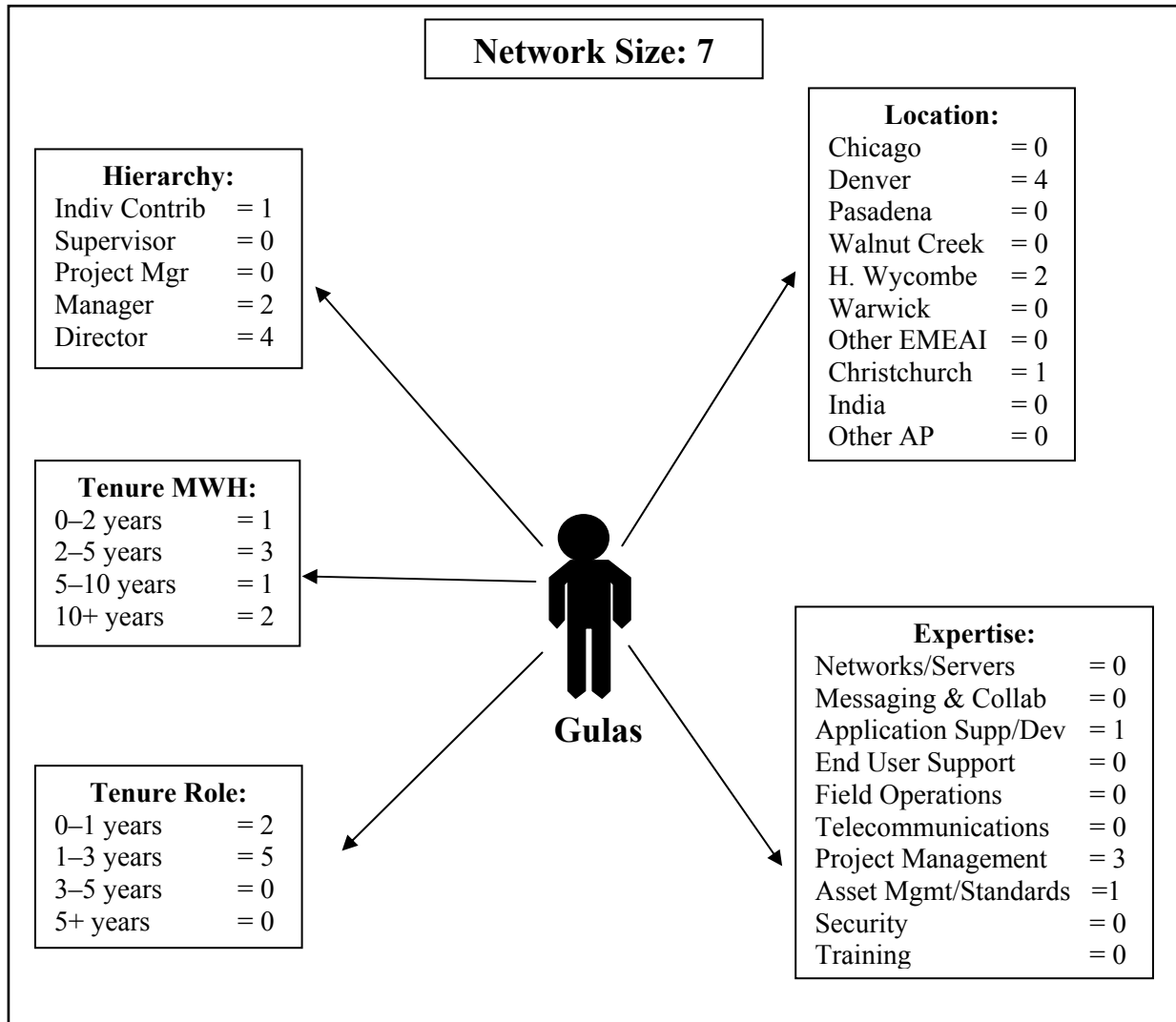


Source: Created by case writer.

Exhibit 8

**BUILDING A NETWORKED ORGANIZATION:
RESTRUCTURING THE IT DEPARTMENT AT MWH (A)**

Vic Gulas's Personal Network Profile
(numbers reflect quantity of relationships in each category)



Source: Created by case writer.

Appendix

**BUILDING A NETWORKED ORGANIZATION:
RESTRUCTURING THE IT DEPARTMENT AT MWH (A)**

Business Goals	Action	Required Connectivity
Reduce costs, standardize services.	Set up global service center in New Zealand.	• Establish new links between people transferring to New Zealand and employees already there. • Lessen reliance on some established relationships that are no longer necessary. • Foster healthy collaboration between service center and all other MWH operations, particularly Denver, Pasadena, High Wycombe, and Warrington.
Reduce costs, standardize services.	Set up Resource Net in India.	• Strengthen ties to the Denver office, other sites in the Americas, and to sites in Europe.
Standardize services, increase innovation.	Take initial steps toward developing a global ERP system by upgrading the system used in the Americas.	• Create productive ties between the ERP team in Pasadena and other MWH sites, particularly Denver.
Standardize services, improve service to internal and external clients, increase ability to innovate.	Establish expertise groups.	• Establish information-sharing ties among people who perform the same kind of work, especially across geographical areas.

Source: Created by case writer.